Social-Media-as-Predictive-Information-for-Stock-Market

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Abstract

Financial social media like StockTwits holds rich discussion on public companies, with users sharing opinions, news reactions, and optional Bullish/Bearish labels. This text enables two key research questions: Can NLP models accurately infer an investor's market stance from their post? And does this sentiment predict future stock performance?

Introduction

The stocktwits article analysed the message sentiment with bow and TF-IDF. Then it tooks the generated sentiment to answer if the price evolution is predictible using a simple model. We choose to test and reproduce this approach, and we go further to really try to predict the market.

1 Prediction of Stocktwits messages sentiments

1.1 Logistic Regression Baseline: Methodology and Lexical Interpretability

1.1.1 Logistic Regression Model Chosen for Its Explainability

Logistic regression is one possible model for classifying the sentiment of Stocktwits messages. The advantage of this model compared with models that are more complex in their architecture, such as Transformers, is explainability (XAI).

1.1.2 Corpus Preprocessing and TF-IDF Feature Representation

The data we used to train and test our model consists of 100,000 messages, the sentiment of which was manually annotated. They were randomly extracted from Stocktwits messages such that this data were composed to a 50/50 split[1] between bullish messages and bearish one. We tokenised and lemmatised these messages using spaCy. The pipeline identifies cashtags (e.g. "$AAPL") and ignores them[2]. It also removes punctuation, spaces and stop words, then normalises each term to lower case and its lemma.

The representation we have chosen is TF-IDF. We preferred this to word2vec because, in short messages, decisions often depend on specific terms. TF-IDF will assign a high weight to these words, whereas word2vec will produce a vector that averages the semantics of each word in a message. Moreover, word2vec's dense 200-dimensional vector space does not allow for the same level of interpretability as the TF-IDF vocabulary. We have also chosen to include bigrams in our vocabulary rather than just simple monograms. In this way, we preserve the semantics of certain expressions. This greatly increases the size of our vocabulary; however, to counter this expansion, we drop terms that are either too rare or too frequent[3].

1.1.3 Regularization-Driven Feature Selection and Sentiment Interpretability

We tested several types of regularisation (ridge, Lasso, ElasticNet) to improve the performance of our logistic regression. Tje best one was ElasticNet regularisation, with an F1-score of 0.73 We find almost the same result for the F1-score calculated per class (bullish (0.75) and bearish (0.72)), which is a good news about the validity of the overall F1-score. Thanks to this regularisation, we find that almost half of the words in our vocabulary ($\approx 16,000$) are not explanatory (parameters/log odds ratio = 0). We can therefore reduce our vocabulary to $\approx 8,000$ words. So what about this list, and what does it tell us about the nature of our messages ?

[1] otherwise, our classifier would have been biased to the point of systematically predicting the predominant (bullish) sentiment in our training dataset

[2] to prevent our classifier from being biased by stocks with very long-term trend

[3] words appearing fewer than 5 times in our corpus (too rare), as well as words appearing in more than 85% of our corpus (too common), are excluded from the vocabulary

- When we look at the most explanatory words in our regression for 'bullish' (Figure.2) (corresponding to the largest parameters/log odds ratios in our model), the lexical field shows a very clear bullish and speculative tendency. There is a many words associated with expansion (explode, soar, accumulation, boom, runner) and encouragement to buy (load, add, undervalue), reinforced by emojis of optimism representing growth. There are also instructions (don't sell, load, bear trap) aimed at maintaining the buying trend.

- When we look at the most explanatory terms in our regression for "bearish" (Figure.3) (corresponding to the largest parameters/log odds in our model), the vast majority of terms are highly negative and serve to disqualify the assets (garbage, trash, scam, trap, fall, junk, disaster, terrible, dead). Moreover, emojis representing collapse and death, reinforce this pessimistic sentiment. We also find instructions (add short, easy short, great short, hold short, sell signal) that are heavily oriented towards short selling.

In conclusion, our logistic regression model – used as a classifier for the sentiment of Stocktwits messages represented by TF-IDF – may not be the most effective. However, its interpretability makes it useful when seeking to understand the nature of the most bullish or bearish messages among our corpus.

1.2 Training Sentiment Labelling with Transformers

1.2.1 Data preparation and selection

For the study, we took all the messages and kept those annotated, I balanced the number of BULLISH/BEARISH for training and testing. The messages were modified to remove any mention of their ticker in order not to learn the popularity of the symbols but rather the meaning of the sentences. The tickers considered are TSLA, AAPL, AMD, NVDA, GOOGL, META : chosen because users who talk about one will talk about the others and that helps to generalize when creating a user embedding to add context for the prediction.

Table 1: Model used / number of messages used / accuracy on balanced dataset

model used	10k	100k	400k	1M	2M
microsoft/mpnet-base	71.8%	70.5%	83.0%	coll.	81.7%
distilroberta-base	69.1%	77.7%	77.9%	–	–
albert-large-v2	70.5%	72.6%	–	–	–
google/electra-base-discriminator	71.8%	78.9%	79.8%	81.3%	82.4%
google/electra-small-discriminator	61.1%	72.9%	72.8%	–	–
cardiffnlp/twitter-roberta-base-sentiment	70.5%	80.7%	82.2%	82.3%	82.9%
cardiffnlp/twitter-roberta-base	71.8%	78.5%	80.8%	79.6%	82.9%
yiyanghkust/finbert-tone	66.4%	75.4%	77.0%	79.3%	82.0%

After balancing, about 80% of the messages are removed (so 100_000 corresponds to about 20_000 actually used)

Table 2: Accuracy on balanced dataset depending on the number of tokens in the message

T≤32	32<T≤64	64<T≤128	128<T≤256
82.5%	88.0%	91.9%	91.1%

Training on 1500000 messages and testing on 15000 tweets.

For the study, we used a bounded token list to allow a huge gpu performance gain and lower memory usage : first tweets under 32 then between 32 and 64 (less numerous) etc. To interpret a generated embedding, the best was to retrieve a layer of the model rather (and add a simple layer) than making a fully connected network. One must especially not train the transformer for more than two epochs otherwise it learns by heart: we see the loss decrease but the performance does not increase (or even decreases). The solution is to use many messages (which is quite slow to process).

1.2.2 Adding context to improve prediction

We used 1500000 messages for training (for RAM reasons)

2 ideas:

- Some users have their own way of expressing themselves and it is interesting to give a description of the user: we can represent it as the average of their previous embeddings (all tickers combined). The embedding thus obtained is reduced to a latent of dimension 2 for generalization.
- Some messages are neutral and come in response to others; some are ambiguous: A first idea consists in taking into account the previous messages to see if we can clarify these tweets: each tweet produces an embedding and we create a previous column which is a weighted average of the previous tweet with a basic attention mechanism: `current_mean := (previous_embedding + current_mean) / 2`
the embedding thus obtained is also reduced to a latent of dimension 2 for generalization.

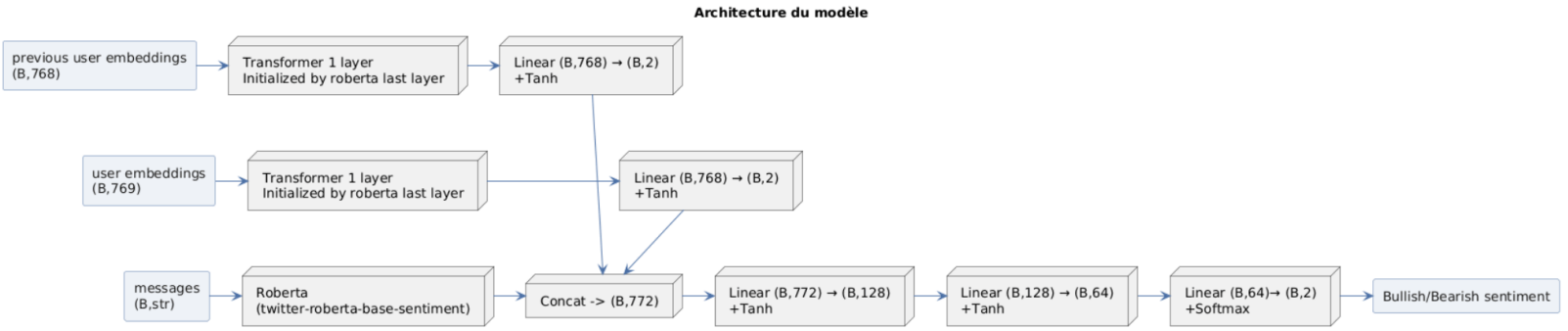


Figure 1: General diagram of the model with previous embedding and user embedding

Performances evaluated on 60000 tweets

Model used	performances
roberta + simple classification head	83.14%
adding user embedding	88.24%
adding previous embedding	86.19%
adding previous embedding and user embedding	88.72%

Performances evaluated on 15000 tweets, by number of tokens

	T≤32	32<T≤64	64<T≤128	128<T≤256
roberta + simple classification head	82.5%	88.0%	91.9%	91.1%
adding user embedding	88.9%	91.4%	89.4%	91.1%
adding previous embedding	86.6%	86.7%	88.6%	90.2%
adding previous embedding and user embedding	89.1%	89.6%	88.7%	90.3%

Context give a significant gain for short messages (¡64 for user embedding, ¡32 for previous embedding). On long tweets (which are much less numerous), we note a slight drop in performance. But the overall performance improves. If we took the max performances for each model, we see that we would reach 90% overall ...

1.2.3 Conclusion for labelling sentiment

We can therefore have a very good prediction of tweets by taking into account the personality of the user and their previous tweets, even with ambiguous tweets. Adding context is a good idea to improve overall performances ...

1.2.4 Remaining points

- It would be interesting to provide the number of tokens to the model so that it can learn to trust more the user embedding when it has enough tokens (and therefore use less the context information) : With a little more time on this project, this would have been the next step.
- We could try to improve the handling of previous messages: tune the attention better
- Handle tweets that are replies, add the initial tweet in context
- It would be interesting to see if a user embedding remains credible for a long time if we do not update it
- The hyperparameters of the created network should be optimizable

- Sometimes a model "collapses" at the beginning of training (even the simplest model with 1 classification head): In that case, restart training and the shuffle does the job: I think that to stabilize learning it might be interesting to feed it at the beginning with tweets that are certainly non ambiguous.

2 Predicting TSLA Price Movement Using StockTwits Sentiment

We chose to work with data related to Tesla (TSLA) stock. This decision was motivated by the fact that it attracts significant interest from investors. As a result, many posts about this stock have been shared on Stocktwits, and this happens regularly. Hence, Tesla (TSLA) stock is an ideal candidate for training our machine learning model, which aims to predict its price movement over the next x days.

2.1 Data Overview and Sentiment Labeling

Our initial data takes the form of a table named "df_tsla", with 5548219 rows, each corresponding to a message posted by one of the 252990 users discussing the topic during the period from 26 May 2008 to 30 December 2023. Over this period, there were 4699 days of activity – in other words, 4699 days on which at least one message was posted about Tesla on Stocktwits. For each of these messages, we know, at the D+1, D+3, D+7 and D+30 time horizons, if Tesla's stock price had risen or fallen since the day of publication. The price of Tesla's stock on a given date is determined at market close. The bullish or bearish sentiment of $\approx 46\%$ of these messages was annotated manually. For the rest of the messages, we chose to infer their sentiment[4] using the RoBERTa + simple classification head model that we had previously trained 1.2.2.

2.2 Filtering Noise: Identification of Tesla Community Experts

Initially, we examined the reliability of all these messages at the D+1, D+3, D+7 and D+30 time horizons, by comparing their sentiments with the actual movements in the share price. The results are poor; regardless of the time horizon chosen, the reliability of sentiment across all 5,548,219 messages ($\approx 50\%$) is not better than a coin flip. However, this result had already been described in [XL06]

So is it possible that, within this white noise represented by the sentiments expressed in these 5,548,219 messages, an informative signal might emerge ? To answer this question, we have chosen to apply a rigorous filtering process to our data. We restrict our sample to a subset of Stocktwits users meeting two criteria: an average reliability score above the 60th quantile and an activity on the platform exceeding 90 days. This reduces the list of users from $252,990$ to 56. We define them as "Tesla community experts". Among these experts, we observe a median activity period of 139 days, with a median reliability at day $D+1$ of $\approx 65\%$, at $D+3$ of $\approx 67\%$, at $D+7$ of $\approx 69\%$ and at $D+30$ of $\approx 74\%$

At this stage, we have to know that the activity of our expert community is not constant over time; we can identify two periods around the month of august in the year 2022[5]. Moreover, when we calculate the 90-day moving average reliability for the entire expert community, we observe that this is not constant either. We will see later that these two factors will play a decisive role when our classifier assigns weights to each expert in this community.

2.3 Experts Validation and Classifier Specification

2.3.1 Statistical Significance of Experts Reliability

Before building our classifier, we examined the significance of our filtered data. In particular, we looked at the significance of the reliability of each of our experts at the $D+1$, $D+3$, $D+7$ and $D+30$ time horizons. To do this, we carried out a one-sided binomial test with the null hypothesis H_0: "The expert's true success rate is $(p = 0.5)$", and therefore the alternative hypothesis H_1: "The expert's success rate is $(p > 0.5)$". Following the tests, the largest p-value[6] is $\approx 9.6 \cdot 10^{-6} \ll 0.05$. Thus, all 56 experts have a p-value pretty smaller than 0.05, which unambiguously rejects the null hypothesis H_0 that their reliability could be due solely to chance. Our community of experts therefore appears to be statistically reliable.

2.3.2 Classifier Architecture and Regularization Tuning

To build our classifier, we chose logistic regression. Our model will be trained to weight the sentiments of our Tesla community experts (explanatory variables X) according to whether the Tesla (TSLA) share price rises or falls at a

[4] Although the RoBERTa + simple classification head model model's accuracy is not 100% (in this case, $\approx 83.14\%$), we will use these predicted sentiments as ground truth to train our classifier.

[5] This increase in the volume of messages around August 2022 can be attributed to the combination of the 3-for-1 share split, which took effect on the 25th August 2022 and market speculation caused by Elon Musk's massive sale of Tesla's shares to finance the acquisition of Twitter.

[6] p-value $\approx 9.6 \cdot 10^{-6}$ was calculated for the expert with user_id = 1716381

30-day horizon. We chose this time horizon because we had previously observed that this was where our experts were most reliable.

To refine our model, we tested three regularisation methods: ridge, Lasso and ElasticNet. For each of these three regularisation methods, we carried out cross-validation to determine the best hyperparameter(s). These cross-validations produce almost identical results for all three types of regularisation, with an F1-score of = 0.70. We therefore opted for ridge regularisation as it handles multicollinearity very well; we will see later that this may be present in our data.

2.4 Model Interpretation and Feature Importance Analysis

2.4.1 Positivity of Classifier Coefficients

We observe that all the coefficients of our logistic regression (l2 ridge regularised with $\lambda = 79$) associated with the 56 experts from the Tesla community are strictly positive $\forall d \in \{\text{Tesla community expert}\}$, $0.03 < \beta_d < 0.35$. Thus, as predicted by the filtering we carried out previously on the initial ≈ 250000 users, none of the experts we identified influences the model in a direction contrary to their sentiment.

2.4.2 Discrepancy between Average Reliability and Feature Weighting

Now, if we consider the hierarchy according to which our logistic regression model weights our experts' opinions (the coefficients β_d), we note that it does not systematically rely on the experts who are, on average, the most reliable[7] when making its decisions. Let us consider two of the most revealing examples :

Let's start by looking at the examples of Expert N°1 (user_id = 2157044) and Expert N°9 (user_id = 5987135) according to this ranking. For Expert N°1, his reliability over a 30-day period is (≈ 0.81), based on his 172 days of activity (the average activity period for the expert community being 175 days). However, Expert No. 9 (user_id = 5987135), over his 162 days of activity, is reliable in (≈ 0.83) of cases. At first, these two experts appear to be on a par. Nevertheless, Expert N°1 was active during a period when the overall activity of the expert community was low (figure.5), whereas Expert N°9 was active during a period of high activity (figure.6).

Now, let us take the example of Expert N°54 (user_id = 5987135). This expert is the most active in the community, with a total of 799 days of activity, and his reliability (across all periods) at a 30-day horizon is (≈ 0.74). However, when we examine their reliability during periods when the overall activity of the expert community was low, their reliability at a 30-day horizon (≈ 0.50) drops dramatically to the point where they become as reliable as a "coin flip". One might therefore wonder if this user is more of a "follower" than an expert. After all, an expert is capable of remaining relevant at times when few others are able to take a stance, is he not?

2.4.3 Information Richness and Multicollinearity Mitigation

Thus, our logistic regression not only weights our experts' opinions according to their reliability and assiduity, but also according to the additional information they provide compared with their peers. One might call this the "richness" of their parole. This is further reinforced by L2 (ridge) regularisation. Indeed, as well as avoiding the risk of overfitting, this regularisation deals with issues of multicollinearity. Yet our data, particularly during periods of high activity, can tend towards collinearity. Experts tend to align their views, and may even influence one another. This results in a reduction, for each of them, in the richness of the information they provide and, consequently, in the value of the coefficient β_d for $d \in \{\text{Tesla community expert}\}$

2.5 Perspectives and Limitations of this model predicting share price movements at the 30-day time horizon

Although the results of our model are fairly good, we must keep in mind that its predictive power is entirely dependent on the future activity of our experts. 'The more of them who contribute on the same day, the more reliable the prediction will be.' Unfortunately, they are few in number (56 experts selected from an initial pool of approximately 250000 users). Yet, this model is merely a proof of concept that would benefit from being expanded into a more general framework, in which a community of experts would be defined and maintained. This community would be much larger, as it would not depend only on the Tesla's share price market. Consequently, given the significantly higher global activity within this community, we could opt for logistic regression with a reject option. A safety measure, in order to manage risk at the time of investment, maybe.

Conclusion

This study analysed the predictive potential of financial social media by examining millions of StockTwits messages in two stages: first, predicting the sentiment of these posts, and second, forecasting share price movements based

[7] the correlation between reliability and ≈ 0.46

on this sentiment information.

Firstly, we demonstrated that although logistic regression with TF-IDF offers a good lexical explainability, Transformer architectures (notably RoBERTa) achieve greater accuracy. Moreover, incorporating user context (embeddings) and message history significantly improves the classification of ambiguous messages and short messages. This is a very good news, since these messages are tweets.

Secondly, if we consider the sentiment of messages from all users, we obtain noise that is completely unusable for training a model (equivalent to coin flips). However, rigorous filtering of activity and reliability allowed us to identify 56 'Tesla community experts'. Applying a regularised (Ridge) logistic regression to these experts enables us to predict share price movements at the 30-day time horizon with an F1 score of 0.70. Analysis of the coefficients reveals that the model prioritises the informational richness of the experts during periods of low consensus, rather than simply their average success rate.

References

[XL06] Aaron Kaufman Xingji Li, Nasser Al Ansari. Stocktwits: Comprehensive records of a financial social media platform from 2008 to 2022. *JQD:DM*, 5:1–41, 2025-10-06.

A Visualizations and Graphs

A.1 Figures about : "Logistic Regression Baseline: Methodology and Lexical Interpretability"

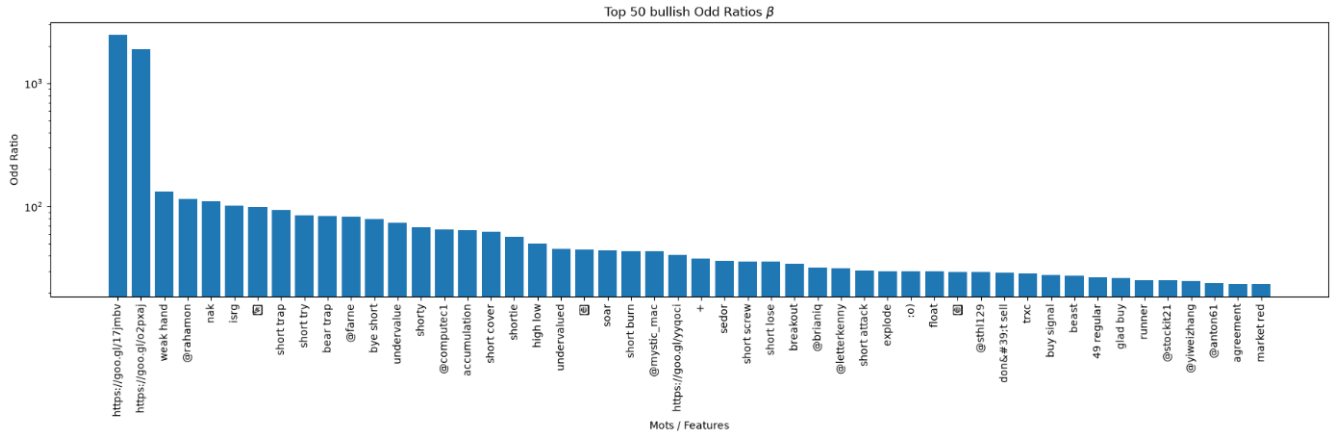


Figure 2: Top 50 bullish Odd Ratios

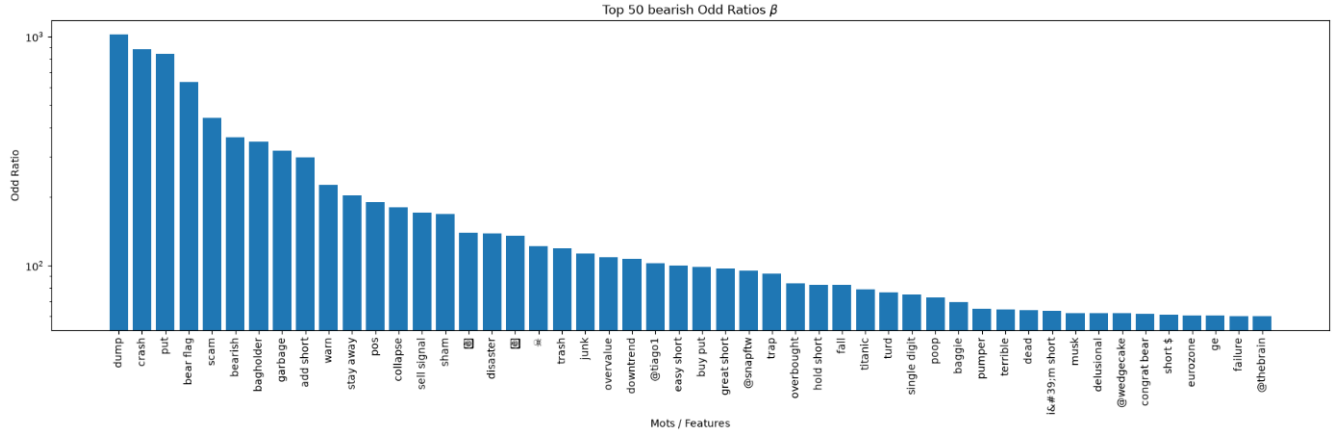


Figure 3: Top 50 bearish Odd Ratios

A.2 Figures about : "Predicting TSLA Price Movement Using StockTwits Sentiment"

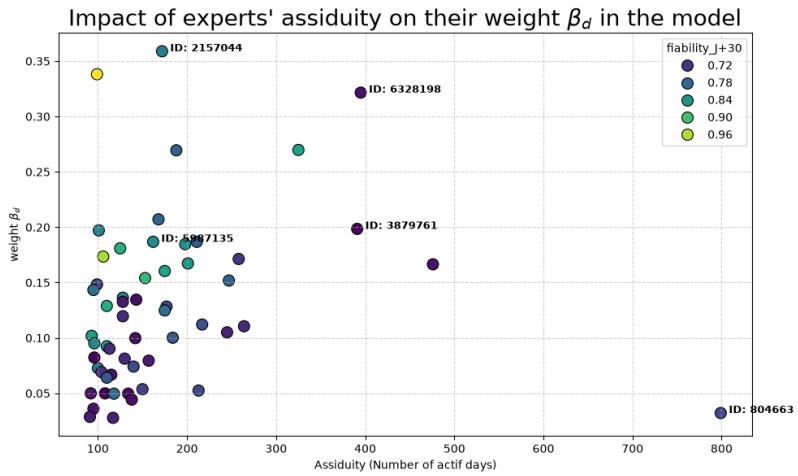


Figure 4: Impact of experts' assiduity on their weight β_d in the model

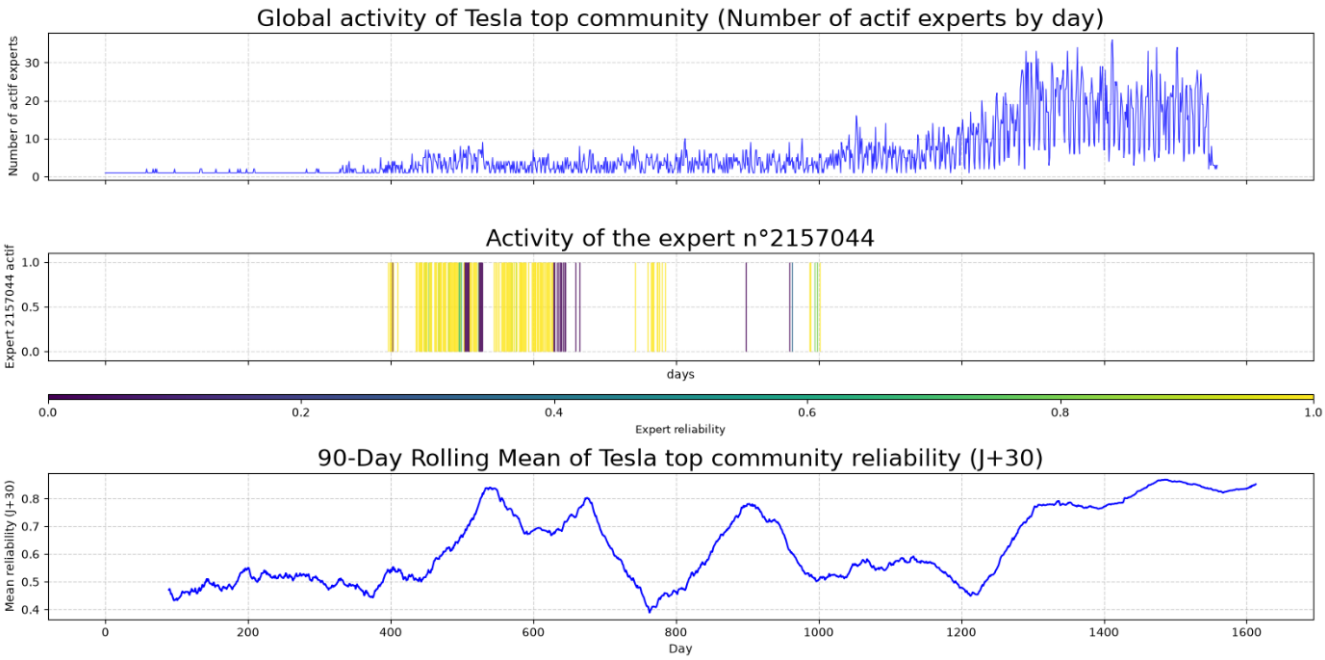


Figure 5: Analysis of activity and reliability of user N°2157044

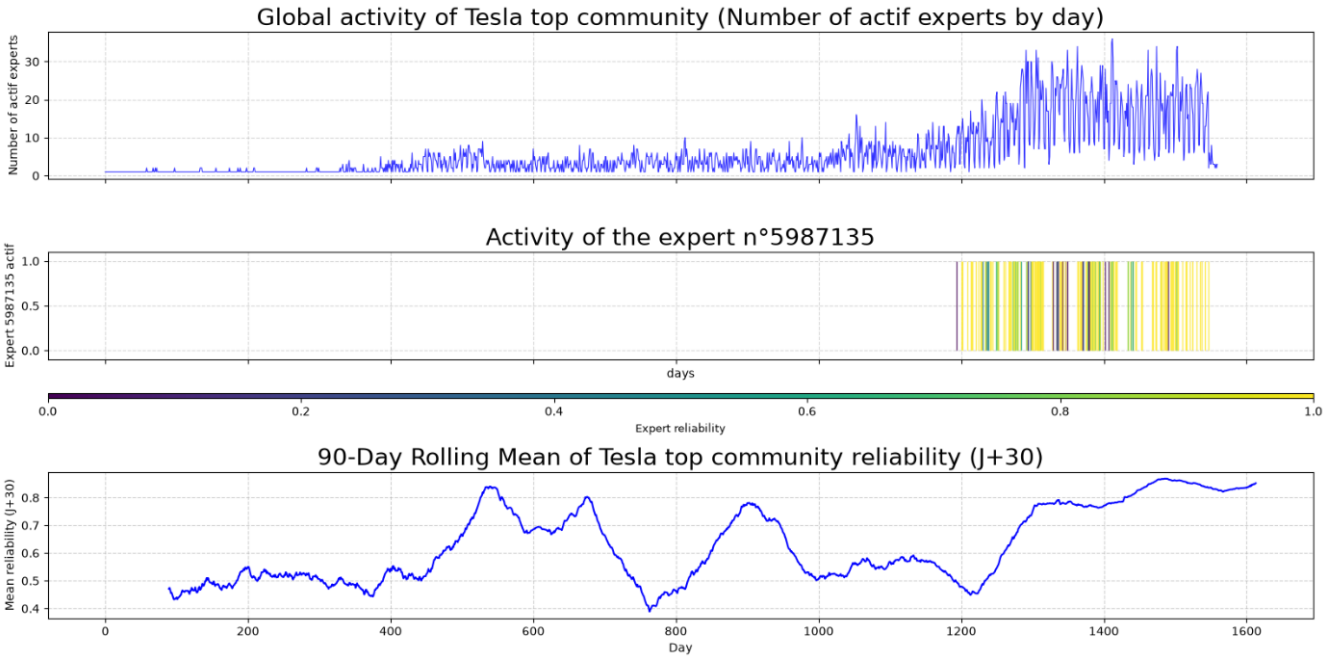


Figure 6: Analysis of activity and reliability of user N°5987135

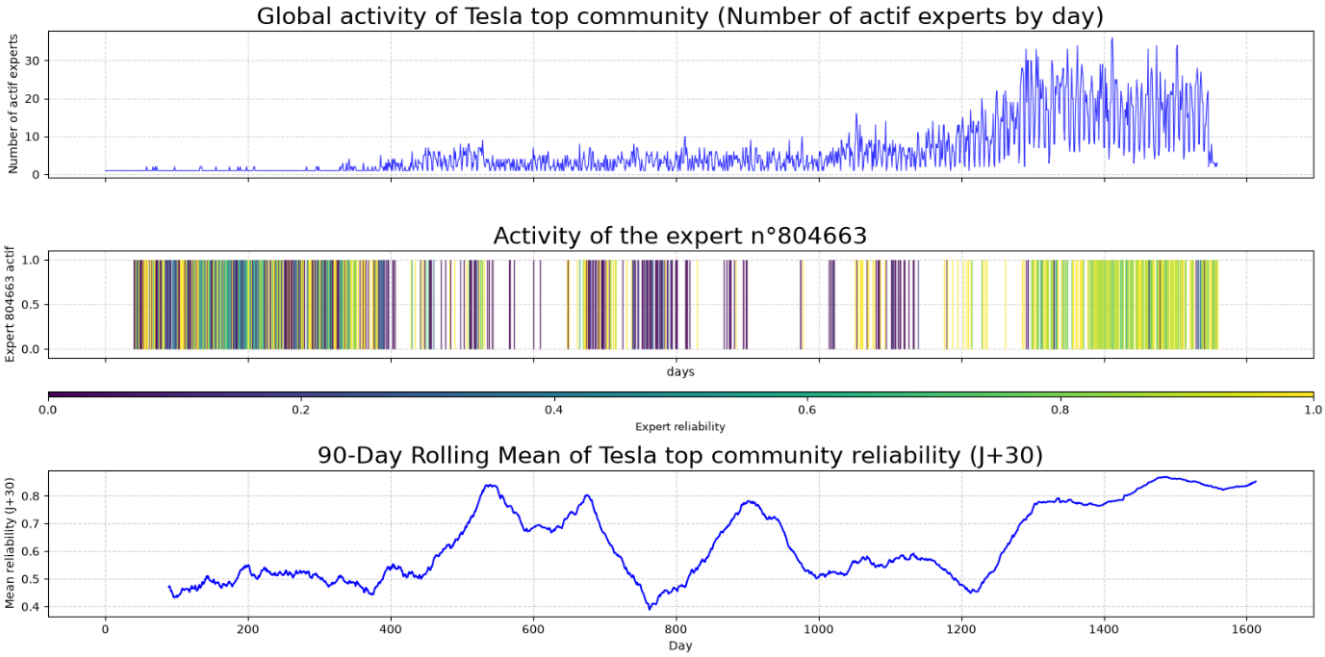


Figure 7: Analysis of activity and reliability of user N°804663